

APPLIED MATHEMATICS

Fourier Series & Partial Differential Equations

Comprehensive Study Plan with MCQs & PSC Tips

PART I: Fourier Series

- Trigonometric Fourier Series
- Bessel Inequality
- Summation of Infinite Series
- Gibbs Phenomenon
- Fourier Sine & Cosine Series
- Parseval's Identity
- Convergence (Dirichlet Conditions)
- Half-Range Expansions

PART II: Partial Differential Equations

- First-Order PDEs (Lagrange's Method)
- Separation of Variables
- Laplace Equation ($\nabla^2 u = 0$)
- Heat / Diffusion Equation ($u_t = ku_{xx}$)
- Classification of 2nd-Order PDEs
- Boundary Value Problems
- Wave Equation ($u_{tt} = c^2 u_{xx}$)
- D'Alembert's Solution

Each section: **Core Concept** → **Key Formulas** → **Worked MCQs + Full Explanations** → **PSC Exam Tips**

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PART I: Fourier Series

Trigonometric Series · Sine & Cosine · Bessel Inequality ·
 Convergence · Infinite Sums

1. Trigonometric Fourier Series

Core Concepts

Any piecewise smooth periodic function $f(x)$ with period $2L$ can be written:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right]$$

Euler–Fourier coefficients:

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx, \quad a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx, \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx$$

Complex (exponential) form: $f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L}, \quad c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-in\pi x/L} dx$

Key facts: Even function $\Rightarrow b_n = 0$ (pure cosine series); Odd function $\Rightarrow a_n = 0$ (pure sine series).

★ Key Formulas & Results ★

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx, \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx$$

Orthogonality (period $2L$):

$$\int_{-L}^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = L\delta_{mn} \quad (m, n \geq 1); \quad \int_{-L}^L \sin \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx = L\delta_{mn}$$

$$\int_{-L}^L \cos \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx = 0 \quad (\text{all } m, n)$$

$$c_n = \frac{1}{2}(a_n - ib_n), \quad c_{-n} = \frac{1}{2}(a_n + ib_n), \quad c_0 = a_0/2$$

Q1.1. The Fourier series of $f(x) = x$ on $(-\pi, \pi)$ (period 2π , $L = \pi$) is:

(A) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \cos nx$

(B) $2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx$

(C) $\frac{\pi}{4} + \sum_{n=1}^{\infty} \frac{1}{n} \sin nx$

(D) $\sum_{n=1}^{\infty} \frac{1}{n} \sin nx$

Answer & Full Explanation

Correct Answer: (B)

$f(x) = x$ is **odd** $\Rightarrow a_0 = a_n = 0$.

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} x \sin nx \, dx. \quad \text{Integrate by parts:} =$$

$$\frac{2}{\pi} \left[-\frac{x \cos nx}{n} \Big|_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos nx \, dx \right] = \frac{2}{\pi} \cdot \frac{-\pi(-1)^n}{n} = \frac{2(-1)^{n+1}}{n}.$$

$$f(x) = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx.$$

Q1.2. The Fourier series of $f(x) = |x|$ on $(-\pi, \pi)$ (period 2π) is:

(A) $\frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2((-1)^n - 1)}{n^2\pi} \cos nx$

(B) $\sum_{n=1}^{\infty} \frac{4}{(2n-1)^2\pi} \cos nx$

(C) $\frac{\pi}{2} - \sum_{n=1}^{\infty} \frac{4}{(2n-1)^2\pi} \cos(2n-1)x$

(D) $\frac{\pi}{2} + \sum_{n \text{ odd}} \frac{4}{n^2\pi} \cos nx$

Answer & Full Explanation

Correct Answer: (C)

$f(x) = |x|$ is **even** $\Rightarrow b_n = 0$.

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} |x| \, dx = \frac{2}{\pi} \int_0^{\pi} x \, dx = \pi.$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \cos nx \, dx = \frac{2}{\pi} \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi} = \frac{2}{n^2\pi} ((-1)^n - 1).$$

For even n : $a_n = 0$. For odd $n = 2k - 1$: $a_n = -\frac{4}{n^2\pi}$.

$$|x| = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\cos(2k-1)x}{(2k-1)^2}.$$

Q1.3. At a jump discontinuity $x = x_0$, the Fourier series of a piecewise smooth function converges to:

- (A) $f(x_0)$
- (B) $f(x_0^+)$
- (C) $\frac{1}{2}[f(x_0^+) + f(x_0^-)]$
- (D) 0

Answer & Full Explanation

Correct Answer: (C) $\frac{1}{2}[f(x_0^+) + f(x_0^-)]$

Dirichlet's theorem: At a jump discontinuity the Fourier series converges to the **average** of the left- and right-hand limits.

PSC classic: For $f(x) = x$ on $(-\pi, \pi)$, at $x = \pm\pi$ the series converges to $\frac{1}{2}[\pi + (-\pi)] = 0$.

Q1.4. The Fourier series of $f(x) = x^2$ on $(-\pi, \pi)$ is $\frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx$. Setting

$x = \pi$ yields $\sum_{n=1}^{\infty} \frac{1}{n^2} =$:

- (A) $\frac{\pi^2}{3}$
- (B) $\frac{\pi^2}{6}$
- (C) $\frac{\pi^2}{12}$
- (D) $\frac{\pi^2}{8}$

Answer & Full Explanation

Correct Answer: (B) $\frac{\pi^2}{6}$

At $x = \pi$ (continuity): $\pi^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n(-1)^n}{n^2} = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{1}{n^2}$.

$$4 \sum_{n=1}^{\infty} \frac{1}{n^2} = \pi^2 - \frac{\pi^2}{3} = \frac{2\pi^2}{3} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

This is the **Basel problem** result — one of the most tested Fourier series applications in PSC exams.

Q1.5. Which of the following is an even function on $(-\pi, \pi)$?

- (A) $f(x) = x \sin x$

- (B) $f(x) = \sin x$
 (C) $f(x) = x^3$
 (D) $f(x) = x \cos x^2 + x$

Answer & Full Explanation

Correct Answer: (A) $f(x) = x \sin x$

$f(-x) = (-x) \sin(-x) = (-x)(-\sin x) = x \sin x = f(x)$. **Even.**

(B) $\sin(-x) = -\sin x$: odd. (C) $(-x)^3 = -x^3$: odd. (D) $(-x) \cos(-x)^2 + (-x) = -x \cos x^2 - x \neq f(x)$: odd.

Rule: (even) $\times$ (even) = even; (odd) $\times$ (odd) = odd; (even) $\times$ (odd) = odd.

2. Fourier Sine and Cosine Series; Half-Range Expansions

Core Concepts

For a function f defined on $(0, L)$ we can form:

Half-range Fourier cosine series (extend as even function): $f(x) = \frac{a_0}{2} +$

$$\sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L}, \quad a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx$$

Half-range Fourier sine series (extend as odd function): $f(x) =$

$$\sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}, \quad b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

Both representations converge to $f(x)$ on $(0, L)$ at points of continuity.

★ Key Formulas & Results ★

Fourier sine coeff: $b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$

Fourier cosine coeff: $a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx$

Standard integrals: $\int_0^L \sin^2 \frac{n\pi x}{L} dx = \frac{L}{2}; \quad \int_0^L \cos^2 \frac{n\pi x}{L} dx = \frac{L}{2}$

$$b_n \text{ for } f(x) = 1 \text{ on } (0, L) : b_n = \frac{2}{n\pi} (1 - (-1)^n) = \begin{cases} 4/(n\pi) & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

Q2.1. The Fourier sine series of $f(x) = 1$ on $(0, \pi)$ is:

- (A) $\frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\sin(2n-1)x}{2n-1}$
- (B) $\sum_{n=1}^{\infty} \frac{\sin nx}{n}$
- (C) $\frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1-(-1)^n}{n} \sin nx$
- (D) Both (A) and (C) are correct

Answer & Full Explanation

Correct Answer: (D) Both (A) and (C) are correct

$$b_n = \frac{2}{\pi} \int_0^{\pi} \sin nx \, dx = \frac{2}{\pi} \left[-\frac{\cos nx}{n} \right]_0^{\pi} = \frac{2(1-(-1)^n)}{n\pi}.$$

Odd n : $b_n = 4/(n\pi)$; even n : $b_n = 0$. Writing $n = 2k-1$: $f(x) = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\sin(2k-1)x}{2k-1}$, which is option (A). Option (C) is the same written out before separating even/odd terms.

Q2.2. The Fourier cosine series of $f(x) = x$ on $(0, L)$ has a_0 equal to:

- (A) L
- (B) $L/2$
- (C) $2L$
- (D) L^2

Answer & Full Explanation

Correct Answer: (A) L

$$a_0 = \frac{2}{L} \int_0^L x \, dx = \frac{2}{L} \cdot \frac{L^2}{2} = L.$$

Hence the constant term in the cosine series is $a_0/2 = L/2$, which is the mean value of $f(x) = x$ on $(0, L)$.

Q2.3. The Fourier **cosine** series of $f(x) = x$ on $(0, \pi)$ has a_n ($n \geq 1$) equal to:

- (A) $\frac{2((-1)^n - 1)}{n^2\pi}$
- (B) $\frac{(-1)^n \cdot 2}{n\pi}$
- (C) $\frac{2((-1)^n - 1)}{n^2\pi} \cdot \pi$
- (D) $\frac{2((-1)^n - 1)}{n^2\pi}$

Answer & Full Explanation**Correct Answer: (D)** $\frac{2((-1)^n - 1)}{n^2\pi}$

$$a_n = \frac{2}{\pi} \int_0^\pi x \cos nx \, dx = \frac{2}{\pi} \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^\pi = \frac{2}{\pi} \cdot \frac{(-1)^n - 1}{n^2} = \frac{2((-1)^n - 1)}{n^2\pi}.$$

Odd n : $a_n = -4/(n^2\pi)$; even n : $a_n = 0$. This gives the full cosine series

$$x = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\cos(2k+1)x}{(2k+1)^2}.$$

3. Bessel Inequality, Parseval Identity & Series Summation**Theorem: Bessel Inequality & Parseval's Identity**For f with Fourier coefficients a_n, b_n on $(-L, L)$:

$$\text{Bessel Inequality: } \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \leq \frac{1}{L} \int_{-L}^L [f(x)]^2 \, dx$$

Parseval's Identity (for piecewise smooth f):

$$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) = \frac{1}{L} \int_{-L}^L [f(x)]^2 \, dx$$

Equality holds (Bessel becomes Parseval) iff the Fourier series converges to f in the **mean-square** sense.**Application:** Parseval's identity provides exact closed-form sums of series such as $\sum 1/n^2$, $\sum 1/n^4$, etc.

★ Key Formulas & Results ★

Parseval: $\frac{1}{L} \int_{-L}^L f^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$

From $f(x) = x^2$ on $(-\pi, \pi)$: $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$

From $f(x) = x$ on $(-\pi, \pi)$: $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$

From $f(x) = |x|$ on $(-\pi, \pi)$: $\sum_{n=0}^{\infty} \frac{1}{(2n+1)^4} = \frac{\pi^4}{96}$

$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$; $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$

Q3.1. Using Parseval's identity applied to $f(x) = x$ on $(-\pi, \pi)$, $\sum_{n=1}^{\infty} \frac{1}{n^2}$ equals:

- (A) $\frac{\pi^2}{4}$
- (B) $\frac{\pi^2}{8}$
- (C) $\frac{\pi^2}{6}$
- (D) $\frac{\pi^2}{3}$

Answer & Full Explanation

Correct Answer: (C) $\frac{\pi^2}{6}$

From $f(x) = x$: $a_n = 0$, $b_n = 2(-1)^{n+1}/n$. $\frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{2\pi^2}{3}$.

Parseval: $\sum_{n=1}^{\infty} b_n^2 = \sum_{n=1}^{\infty} \frac{4}{n^2} = \frac{2\pi^2}{3} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

Q3.2. Using Parseval's identity for $f(x) = x^2$ on $(-\pi, \pi)$, $\sum_{n=1}^{\infty} \frac{1}{n^4}$ equals:

- (A) $\frac{\pi^4}{45}$

(B) $\frac{\pi^4}{90}$

(C) $\frac{\pi^4}{96}$

(D) $\frac{\pi^4}{72}$

Answer & Full Explanation**Correct Answer:** (B) $\frac{\pi^4}{90}$

Fourier series of x^2 : $\frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx$. $a_0 = 2\pi^2/3$, $a_n = 4(-1)^n/n^2$, $b_n = 0$.

Parseval: $\frac{1}{\pi} \int_{-\pi}^{\pi} x^4 dx = \frac{2\pi^4}{5} = \frac{a_0^2}{2} + \sum a_n^2 = \frac{2\pi^4}{9} + 16 \sum \frac{1}{n^4}$.

$$16 \sum \frac{1}{n^4} = \frac{2\pi^4}{5} - \frac{2\pi^4}{9} = \frac{8\pi^4}{45} \Rightarrow \sum \frac{1}{n^4} = \frac{\pi^4}{90}.$$

Q3.3. The Bessel inequality states that the sum of squares of Fourier coefficients is:

(A) Equal to the square of f (B) At most $\frac{1}{L} \int_{-L}^L f^2 dx$ (C) At least $\frac{1}{L} \int_{-L}^L f^2 dx$

(D) Zero

Answer & Full Explanation**Correct Answer:** (B)

$$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \leq \frac{1}{L} \int_{-L}^L f^2 dx.$$

Equality (Parseval) holds iff the Fourier series represents f in the L^2 (mean-square) sense. The Bessel inequality follows from $\|f - S_N\|^2 \geq 0$ where S_N is the N -th partial sum.

4. Convergence of Fourier Series (Dirichlet Conditions)

Theorem: Dirichlet's Theorem

If f is periodic with period $2L$ and on $[-L, L]$:

1. f is piecewise continuous (finitely many discontinuities), and
2. f is piecewise monotone (finitely many turning points),

then the Fourier series converges:

- to $f(x)$ at every point of **continuity**,
- to $\frac{1}{2}[f(x^+) + f(x^-)]$ at every **jump discontinuity**.

Gibbs Phenomenon: Near a jump discontinuity the partial sums overshoot by about 9% of the jump height, regardless of how many terms are taken.

★ Key Formulas & Results ★

Pointwise convergence: $S_N(x) \rightarrow f(x)$ at continuity

At jump: $S_N(x_0) \rightarrow \frac{1}{2}[f(x_0^+) + f(x_0^-)]$

Uniform convergence: if f is continuous and $f(-L) = f(L)$, the series converges uniformly

Mean-square (L^2) convergence always holds for $f \in L^2(-L, L)$

Gibbs overshoot $\approx 9\%$ of jump $\approx 0.18 \int_0^\pi \frac{\sin t}{t} dt - 0.5 \approx 0.0895$

Q4.1. Which of the following ensures pointwise convergence of the Fourier series of f to $f(x)$ at all continuity points?

- (A) f is bounded on $[-\pi, \pi]$
- (B) f is differentiable at x
- (C) f satisfies Dirichlet conditions on $[-\pi, \pi]$
- (D) f is integrable

Answer & Full Explanation

Correct Answer: (C) Dirichlet conditions

Dirichlet conditions (piecewise continuous + piecewise monotone) guarantee pointwise convergence to $f(x)$ at continuity points and to the average at jump discontinuities.

(D) integrability alone (Lebesgue) only guarantees L^2 convergence, not pointwise. (B) differentiability at x gives pointwise convergence at that single point, but Dirichlet gives it everywhere.

Q4.2. The Gibbs phenomenon near a jump discontinuity means that as $N \rightarrow \infty$, the partial sums S_N :

- (A) Overshoot by about 9% and the overshoot disappears
- (B) Overshoot by about 9% of the jump, which does not vanish as $N \rightarrow \infty$
- (C) Converge uniformly near the jump
- (D) Converge to zero near the jump

Answer & Full Explanation

Correct Answer: (B)

The Gibbs phenomenon: near a jump discontinuity, partial sums of the Fourier series **permanently** overshoot (and undershoot) by $\approx 9\%$ of the jump magnitude. This overshoot does not decrease as more terms are added; only its location moves closer to the discontinuity. It is an intrinsic feature of truncated Fourier series.

Q4.3. $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$ (obtained from the Fourier series of the square wave at $x = \pi/2$) equals:

- (A) $\pi/2$
- (B) $\pi/4$
- (C) $\pi/8$
- (D) 1

Answer & Full Explanation

Correct Answer: (B) $\pi/4$

The Fourier series $\frac{4}{\pi} \sum_{n=0}^{\infty} \frac{\sin((2n+1)x)}{2n+1}$ of the square wave $f(x) = \pm 1$ converges to $f(\pi/2) = 1$ at $x = \pi/2$:

$$\frac{4}{\pi} \sum_{n=0}^{\infty} \frac{\sin((2n+1)\pi/2)}{2n+1} = \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = 1.$$

$\Rightarrow \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \frac{\pi}{4}$. This is the **Leibniz–Gregory series**, one of the most famous in mathematics.

Important PSC Exam Points

PSC Must-Know Infinite Series via Fourier Methods:

Series	Value	Source
$\sum_{n=1}^{\infty} \frac{1}{n^2}$	$\frac{\pi^2}{6}$	$f(x) = x$ on $(-\pi, \pi)$
$\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$	$\frac{\pi}{4}$	Square wave at $x = \pi/2$
$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$	$\frac{\pi^2}{8}$	$f(x) = x $ Parseval
$\sum_{n=1}^{\infty} \frac{1}{n^4}$	$\frac{\pi^4}{90}$	$f(x) = x^2$ on $(-\pi, \pi)$
$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$	$\frac{\pi^2}{12}$	$f(x) = x^2$ evaluated at $x = 0$

PART II: Partial Differential Equations

First-Order PDEs · Classification · Separation of Variables ·
 Laplace · Wave · Heat

5. First-Order PDEs: Lagrange's Method

Core Concepts

Quasi-linear first-order PDE: $P(x, y, z)p + Q(x, y, z)q = R(x, y, z)$ where $p = \partial z / \partial x$ and $q = \partial z / \partial y$.

Lagrange's method: Solve the **auxiliary (Lagrange–Charpit) system**:

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

From this system find **two independent first integrals** $u(x, y, z) = c_1$ and $v(x, y, z) = c_2$.

General solution: $F(u, v) = 0$ or $u = f(v)$ for arbitrary function F or f .

Special types: $z_x = f(x)$: integrate directly; $z_x + az_y = 0$: characteristic $y - ax = c$; solution $z = f(y - ax)$.

★ Key Formulas & Results ★

Lagrange's aux. system: $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

General solution: $F(u_1, u_2) = 0$ ($u_1 = c_1$, $u_2 = c_2$ are first integrals)

Transport equation $u_t + cu_x = 0$: solution $u = f(x - ct)$ (wave travelling right)

Linear $z_x + p(x)z = q(x, y)$: treat y as parameter, solve ODE in x

Q5.1. The Lagrange auxiliary system for $yp - xq = z$ is:

(A) $\frac{dx}{y} = \frac{dy}{-x} = \frac{dz}{z}$

(B) $\frac{dx}{-y} = \frac{dy}{x} = \frac{dz}{z}$

(C) $\frac{dx}{y} = \frac{dy}{x} = \frac{dz}{z}$

(D) $\frac{dx}{z} = \frac{dy}{y} = \frac{dz}{-x}$

Answer & Full Explanation

Correct Answer: (A)

$P = y, Q = -x, R = z$. Lagrange's system: $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{y} = \frac{dy}{-x} = \frac{dz}{z}$.

From $\frac{dx}{y} = \frac{dy}{-x}$: $x dx + y dy = 0 \Rightarrow x^2 + y^2 = c_1$. From $\frac{dy}{-x} = \frac{dz}{z}$: further integration gives second integral. General solution: $F(x^2 + y^2, z/\sqrt{x^2 + y^2}) = 0$.

Q5.2. The general solution of $p + q = 1$ (with $p = z_x, q = z_y$) is:

- (A) $z = ax + by + c$ with $a + b = 1$
- (B) $z = f(x - y)$
- (C) $z = x + f(y - x)$
- (D) $z = xy$

Answer & Full Explanation

Correct Answer: (C) $z = x + f(y - x)$

Lagrange: $\frac{dx}{1} = \frac{dy}{1} = \frac{dz}{1}$. First integral from $dx = dy$: $y - x = c_1$. Second integral from $dx = dz$: $z - x = c_2$.

General solution: $z - x = f(y - x)$, i.e. $z = x + f(y - x)$.

Check: $z_x = 1 - f'(y - x), z_y = f'(y - x)$. $z_x + z_y = 1$. ✓

Q5.3. The PDE $\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$ (transport equation) has general solution:

- (A) $u = f(x + ct)$
- (B) $u = f(x - ct)$
- (C) $u = e^{cx} f(t)$
- (D) $u = f(x)g(t)$

Answer & Full Explanation

Correct Answer: (B) $u = f(x - ct)$

Characteristics: $\frac{dt}{1} = \frac{dx}{c} = \frac{du}{0}$.

From $dt = dx/c$: $x - ct = \text{const}$ (characteristics are lines $x = ct + \text{const}$). From $du = 0$: $u = \text{const}$ along characteristics. General solution: $u = f(x - ct)$ — a wave travelling to the **right** with speed c .

6. Classification of Second-Order PDEs

Core Concepts

General second-order PDE: $Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = G$

Discriminant $\Delta = B^2 - 4AC$:

Δ	Type	Canonical Example
< 0	Elliptic	Laplace equation $u_{xx} + u_{yy} = 0$
$= 0$	Parabolic	Heat equation $u_t = ku_{xx}$
> 0	Hyperbolic	Wave equation $u_{tt} = c^2u_{xx}$

Canonical forms: Elliptic $\rightarrow u_{\xi\xi} + u_{\eta\eta} = \Phi$; Parabolic $\rightarrow u_{\xi\xi} = \Phi$; Hyperbolic $\rightarrow u_{\xi\eta} = \Phi$ (or $u_{\xi\xi} - u_{\eta\eta} = \Phi$).

★ Key Formulas & Results ★

$\Delta = B^2 - 4AC < 0$: elliptic; $= 0$: parabolic; > 0 : hyperbolic

Laplace $\nabla^2 u = 0$: $A = C = 1, B = 0, \Delta = -4 < 0$ (elliptic)

Heat $u_t = ku_{xx}$: $A = k, B = 0, C = 0, \Delta = 0$ (parabolic)

Wave $u_{tt} = c^2u_{xx}$: $A = c^2, B = 0, C = -1, \Delta = 4c^2 > 0$ (hyperbolic)

Characteristics: $\frac{dy}{dx} = \frac{B \pm \sqrt{\Delta}}{2A}$

Q6.1. The PDE $u_{xx} + 4u_{xy} + 4u_{yy} = 0$ is classified as:

- (A) Elliptic
- (B) Hyperbolic
- (C) Parabolic
- (D) Mixed

Answer & Full Explanation

Correct Answer: (C) Parabolic

$A = 1, B = 4, C = 4. \Delta = B^2 - 4AC = 16 - 16 = 0 \Rightarrow$ **parabolic**.

Factorisation confirms: $u_{xx} + 4u_{xy} + 4u_{yy} = (\partial_x + 2\partial_y)^2 u = 0$. The single characteristic direction is $dy/dx = 2$.

Q6.2. The classification of $u_{xx} - 2u_{xy} + 5u_{yy} = 0$ is:

- (A) Hyperbolic
- (B) Parabolic
- (C) Elliptic
- (D) Cannot be determined

Answer & Full Explanation**Correct Answer: (C) Elliptic**
 $A = 1, B = -2, C = 5. \Delta = (-2)^2 - 4(1)(5) = 4 - 20 = -16 < 0 \Rightarrow \text{elliptic}.$

Elliptic equations describe steady-state phenomena; no real characteristics exist.

Q6.3. The PDE $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0$ is classified as hyperbolic because:

- (A) $A \neq C$
- (B) $\Delta = B^2 - 4AC = 4 > 0$
- (C) $B = 0$
- (D) The equation is second order

Answer & Full Explanation**Correct Answer: (B) $\Delta = 4 > 0$**
 $A = 1, B = 0, C = -1. \Delta = 0 - 4(1)(-1) = 4 > 0 \Rightarrow \text{hyperbolic}.$

Two real characteristic families: $y \pm x = \text{const.}$ General solution: $u = f(x + y) + g(x - y)$ (d'Alembert type).

7. Method of Separation of Variables

Core Concepts

Assume $u(x, t) = X(x)T(t)$ (or $u(x, y) = X(x)Y(y)$) and substitute into the PDE.

Since the PDE splits into a function of x alone = a function of t alone, both must equal a **separation constant** $-\lambda$ (or λ).

General procedure:

1. Assume $u = XT$ and substitute into PDE.
2. Separate to get $X''/X = T''/T = \text{const} = -\lambda^2$ (or λ).
3. Solve the resulting ODEs for X and T .
4. Apply boundary conditions (BCs) to find eigenvalues λ_n .
5. Apply initial conditions (ICs) to find Fourier coefficients.
6. Write the full solution as a superposition (sum over all n).

★ Key Formulas & Results ★

Sturm–Liouville problem from BCs yields eigenvalues λ_n and eigenfunctions $\phi_n(x)$

Homogeneous BC $X(0) = X(L) = 0$: $\lambda_n = \left(\frac{n\pi}{L}\right)^2$, $X_n = \sin \frac{n\pi x}{L}$

Homogeneous BC $X'(0) = X'(L) = 0$: $\lambda_n = \left(\frac{n\pi}{L}\right)^2$, $X_n = \cos \frac{n\pi x}{L}$

Full solution: $u(x, t) = \sum_n [A_n X_n(x) T_n(t)]$

Coefficients from ICs using orthogonality (Fourier sine/cosine series)

Q7.1. Substituting $u = X(x)T(t)$ into the heat equation $u_t = ku_{xx}$ separates to:

- (A) $T'/T = X''/X = \lambda$
- (B) $T'/(kT) = X''/X = -\lambda^2$
- (C) $T' = kX''$
- (D) $kX''T = XT'$; separation gives same constant on both sides

Answer & Full Explanation

Correct Answer: (B) $T'/(kT) = X''/X = -\lambda^2$

$u_t = XT'$ and $u_{xx} = X''T$. Heat equation: $XT' = kX''T$. Divide by kXT :
 $\frac{T'}{kT} = \frac{X''}{X} = \text{const} = -\lambda^2$.

The sign $-\lambda^2$ is chosen so that X has oscillatory (not exponential) solutions, which are needed for Fourier series. Result: $X'' + \lambda^2 X = 0$ and $T' + k\lambda^2 T = 0$.

Q7.2. For the separated ODE $X'' + \lambda^2 X = 0$ with $X(0) = 0$ and $X(L) = 0$, the eigenvalues λ_n are:

- (A) $\lambda_n = n\pi/L$, $n = 1, 2, 3, \dots$
- (B) $\lambda_n = n^2\pi^2/L^2$
- (C) $\lambda_n = n\pi$, $n = 0, 1, 2, \dots$
- (D) $\lambda_n = 2n\pi/L$

Answer & Full Explanation

Correct Answer: (A) $\lambda_n = n\pi/L$

General solution of $X'' + \lambda^2 X = 0$: $X = A \cos \lambda x + B \sin \lambda x$.

$X(0) = 0 \Rightarrow A = 0$. $X(L) = B \sin \lambda L = 0$. For non-trivial: $\sin \lambda L = 0 \Rightarrow \lambda L = n\pi$. $\lambda_n = \frac{n\pi}{L}$, $n = 1, 2, 3, \dots$

Eigenfunctions: $X_n = \sin(n\pi x/L)$.

8. Laplace Equation: Boundary Value Problems

Core Concepts

Laplace equation (steady-state, elliptic): $\nabla^2 u = u_{xx} + u_{yy} = 0$

Dirichlet problem: specify u on boundary.

Neumann problem: specify $\partial u / \partial n$ (normal derivative) on boundary.

Uniqueness: Solution is unique for Dirichlet; unique up to a constant for Neumann.

Rectangle $[0, a] \times [0, b]$ with $u = 0$ on three sides, $u(x, b) = f(x)$:

$$u(x, y) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{a} \frac{\sinh(n\pi y/a)}{\sinh(n\pi b/a)}$$

where $B_n = \frac{2}{a} \int_0^a f(x) \sin \frac{n\pi x}{a} dx$.

Disk (polar, radius R): $u(r, \theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(\frac{r}{R}\right)^n (a_n \cos n\theta + b_n \sin n\theta)$.

★ Key Formulas & Results ★

$\nabla^2 u = 0$ in rectangle: $u = X(x)Y(y)$;

$$X'' + \lambda^2 X = 0, Y'' - \lambda^2 Y = 0 \Rightarrow Y = A \cosh \lambda y + B \sinh \lambda y$$

Mean value property: $u(x_0, y_0) = \frac{1}{2\pi} \int_0^{2\pi} u(x_0 + r \cos \theta, y_0 + r \sin \theta) d\theta$

Maximum principle: u attains max/min on the boundary, not inside

Polar Laplacian: $\nabla^2 u = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} = 0$

Q8.1. For the Dirichlet problem $\nabla^2 u = 0$ in a rectangle with $u = 0$ on three sides, the Y -equation after separating $u = X(x)Y(y)$ with $X'' + \lambda^2 X = 0$ is:

- (A) $Y'' + \lambda^2 Y = 0$
- (B) $Y'' - \lambda^2 Y = 0$
- (C) $Y' = \lambda Y$
- (D) $Y'' = 0$

Answer & Full Explanation

Correct Answer: (B) $Y'' - \lambda^2 Y = 0$

$$u_{xx} + u_{yy} = 0 \Rightarrow X''Y + XY'' = 0 \Rightarrow \frac{X''}{X} = -\frac{Y''}{Y} = -\lambda^2.$$

$X'' + \lambda^2 X = 0$ (oscillatory) and $Y'' - \lambda^2 Y = 0$ (exponential: $Y = A \sinh \lambda y + B \cosh \lambda y$).

The sign is **opposite** to that of the X -equation — a common PSC trap.

Q8.2. The maximum principle for the Laplace equation states that:

- (A) u can attain its maximum anywhere inside the domain
- (B) The maximum of u is attained on the **boundary** of the domain
- (C) u is always positive
- (D) The minimum of u is zero

Answer & Full Explanation

Correct Answer: (B)

Maximum Principle: A harmonic function ($\nabla^2 u = 0$) in a bounded region D attains its maximum (and minimum) values on the **boundary** ∂D , not in the interior, unless u is identically constant.

Consequence: If $u = 0$ on boundary, then $u \equiv 0$ everywhere (uniqueness of Dirichlet problem).

Q8.3. For Laplace's equation in polar form, a function of r only satisfying $r^2 u_{rr} + r u_r = 0$ has general solution:

- (A) $u = A \ln r + B$
- (B) $u = Ar + B/r$
- (C) $u = Ar^2 + B$
- (D) $u = A \ln r^2 + B$

Answer & Full Explanation

Correct Answer: (A) $u = A \ln r + B$

$$r^2 u_{rr} + r u_r = \frac{d}{dr}(r u') \cdot r = 0 \Rightarrow \frac{d}{dr}(r u_r) = 0 \Rightarrow r u_r = A. \quad u_r = A/r \Rightarrow u = A \ln r + B.$$

This is the 2D Laplace Green's function (fundamental solution). It is not valid at $r = 0$ (singularity).

9. Wave Equation

Core Concepts

1D Wave equation: $u_{tt} = c^2 u_{xx}$ (vibrating string, $c = \sqrt{T/\rho}$)

IBVP (string fixed at both ends): $u(0, t) = u(L, t) = 0$; $u(x, 0) = f(x)$;
 $u_t(x, 0) = g(x)$

Solution by separation: $u(x, t) = \sum_{n=1}^{\infty} \sin \frac{n\pi x}{L} \left(A_n \cos \frac{n\pi ct}{L} + B_n \sin \frac{n\pi ct}{L} \right)$

where $A_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$ and $B_n = \frac{2}{n\pi c} \int_0^L g(x) \sin \frac{n\pi x}{L} dx$.

D'Alembert's solution (infinite string): $u(x, t) = \frac{1}{2}[f(x+ct) + f(x-ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$.

★ Key Formulas & Results ★

$$u_{tt} = c^2 u_{xx}; \quad u = \sum_n \sin \frac{n\pi x}{L} (A_n \cos \omega_n t + B_n \sin \omega_n t), \quad \omega_n = \frac{n\pi c}{L}$$

$$\text{D'Alembert: } u = \frac{1}{2}[f(x+ct) + f(x-ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$$

$$\text{Energy: } E = \frac{1}{2} \int_0^L [\rho u_t^2 + T u_x^2] dx = \text{const}$$

$$\text{Natural frequencies (normal modes): } f_n = \frac{nc}{2L}; \text{ fundamental } f_1 = \frac{c}{2L}$$

Q9.1. The wave equation $u_{tt} = 4u_{xx}$, $0 < x < \pi$, with $u(0, t) = u(\pi, t) = 0$, $u(x, 0) = \sin x$, $u_t(x, 0) = 0$ has solution:

- (A) $u = \sin x \cos 2t$
- (B) $u = \sin x \sin 2t$
- (C) $u = \cos x \cos 2t$
- (D) $u = \sin x \cos t$

Answer & Full Explanation

Correct Answer: (A) $u = \sin x \cos 2t$

$c = 2$, $L = \pi$, $\omega_n = n\pi c/L = 2n$. $f(x) = \sin x = \sin(1 \cdot x)$, $g(x) = 0$.

Only $n = 1$ contributes: $A_1 = 1$ ($\sin x$ coefficient is 1), $B_1 = 0$ (since $g = 0$).
 $u = \sin x \cos(2 \cdot 1 \cdot t) = \sin x \cos 2t$.

Verify: $u_{tt} = -4 \sin x \cos 2t$; $c^2 u_{xx} = 4(-\sin x \cos 2t)$. ✓

Q9.2. D'Alembert's solution $u = \frac{1}{2}[f(x+ct) + f(x-ct)]$ (with $g = 0$) represents:

- (A) A standing wave
- (B) Superposition of two waves: one travelling right, one left
- (C) A wave travelling right only
- (D) An exponentially decaying wave

Answer & Full Explanation

Correct Answer: (B)

$f(x+ct)$ is a wave profile f travelling to the **left** at speed c ; $f(x-ct)$ travels to the **right**. D'Alembert's solution is the superposition of these two waves, each with half the original amplitude.

Physical interpretation: An initial displacement $f(x)$ splits into two waves propagating in opposite directions.

Q9.3. For the wave equation on $[0, L]$ with fixed ends, the fundamental frequency (first normal mode) is:

- (A) c/L
- (B) $c/(2L)$
- (C) $2c/L$
- (D) $\pi c/L$

Answer & Full Explanation

Correct Answer: (B) $c/(2L)$

$\omega_n = n\pi c/L$, so $f_n = \omega_n/(2\pi) = nc/(2L)$. Fundamental: $n = 1$, $f_1 = c/(2L)$.

Overtone: $f_n = nf_1$ (harmonic series). For a guitar string: $c = \sqrt{T/\mu}$ (T = tension, μ = linear density).

10. Heat (Diffusion) Equation

Core Concepts

1D Heat equation: $u_t = k u_{xx}$ ($k > 0$ thermal diffusivity)

IBVP (rod, zero temperature at ends): $u(0, t) = u(L, t) = 0$; $u(x, 0) = f(x)$

Solution: $u(x, t) = \sum_{n=1}^{\infty} B_n e^{-k(n\pi/L)^2 t} \sin \frac{n\pi x}{L}$

where $B_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$.

Key behaviour: The $e^{-k\lambda_n^2 t}$ factor ensures exponential decay; as $t \rightarrow \infty$, $u \rightarrow 0$ (rod cools to zero boundary temperature).

Insulated ends ($u_x(0, t) = u_x(L, t) = 0$): $u(x, t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} A_n e^{-k(n\pi/L)^2 t} \cos \frac{n\pi x}{L}$.

★ Key Formulas & Results ★

$$u_t = ku_{xx}; \quad u = \sum B_n e^{-k\lambda_n^2 t} \sin \frac{n\pi x}{L}, \quad \lambda_n = \frac{n\pi}{L}$$

$$B_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx \quad (\text{Fourier sine coefficients})$$

Steady state ($u_t = 0$): $ku_{xx} = 0 \Rightarrow u = Ax + B$ (linear, from BCs)

Non-homogeneous BCs: subtract steady state first, then apply separation of variables to the homogeneous remainder

Insulated BC: $u_x(0, t) = u_x(L, t) = 0 \Rightarrow$ cosine series; steady state $= a_0/2 = \text{mean IC}$

Q10.1. For the heat equation $u_t = u_{xx}$ on $[0, \pi]$ with $u(0, t) = u(\pi, t) = 0$ and $u(x, 0) = x(\pi - x)$, the solution is:

(A) $u = \sum_{n=1}^{\infty} B_n e^{-n^2 t} \sin nx$

(B) $u = \sum_{n=1}^{\infty} A_n e^{n^2 t} \cos nx$

(C) $u = x(\pi - x)e^{-t}$

(D) $u = \sum_{n=1}^{\infty} B_n e^{-nt} \sin nx$

Answer & Full Explanation

Correct Answer: (A)

$k = 1$, $L = \pi$, eigenvalues $\lambda_n = n$. $T_n(t) = e^{-k\lambda_n^2 t} = e^{-n^2 t}$. $u = \sum B_n e^{-n^2 t} \sin nx$.

$$B_n = \frac{2}{\pi} \int_0^{\pi} x(\pi - x) \sin nx \, dx = \frac{4(1 - (-1)^n)}{n^3 \pi}.$$

$$B_n = \begin{cases} 8/(n^3 \pi) & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

The full solution is $u = \frac{8}{\pi} \sum_{k=0}^{\infty} \frac{e^{-(2k+1)^2 t}}{(2k+1)^3} \sin(2k+1)x$.

Q10.2. As $t \rightarrow \infty$, the solution of the heat equation with homogeneous boundary conditions ($u = 0$ at both ends) approaches:

- (A) $u = u(x, 0)$
- (B) $u = 0$
- (C) $u = \text{steady-state linear function}$
- (D) $u = \text{constant}$

Answer & Full Explanation

Correct Answer: (B) $u = 0$

Each term $B_n e^{-k(n\pi/L)^2 t} \rightarrow 0$ as $t \rightarrow \infty$ (since exponent is negative). The rod cools to the boundary temperature (zero).

With **insulated** BCs ($u_x = 0$ at ends), steady state $= \frac{a_0}{2} = \text{mean initial temperature}$.

With non-zero BCs $u(0, t) = T_1$, $u(L, t) = T_2$: steady state is the linear function $T_1 + (T_2 - T_1)x/L$.

Q10.3. The heat equation $u_t = ku_{xx}$ with insulated ends $u_x(0, t) = u_x(L, t) = 0$ and $u(x, 0) = f(x)$ has steady-state solution:

- (A) $u_\infty = f(0)$
- (B) $u_\infty = \frac{1}{L} \int_0^L f(x) dx$
- (C) $u_\infty = f(L/2)$
- (D) $u_\infty = 0$

Answer & Full Explanation

Correct Answer: (B) $u_\infty = \frac{1}{L} \int_0^L f(x) dx$

With insulated ends, heat is conserved (no flux out). The steady state is a uniform temperature equal to the mean initial temperature: $u_\infty = \frac{1}{L} \int_0^L f(x) dx = \frac{a_0}{2}$ (the constant term in the cosine Fourier series).

Q10.4. The separation constant is chosen negative ($-\lambda^2$) in $X''/X = T'/(kT) = -\lambda^2$ for the heat equation because:

- (A) It makes T oscillate
- (B) It ensures $T \rightarrow 0$ as $t \rightarrow \infty$ (physically correct cooling)
- (C) It simplifies the BCs
- (D) λ^2 must be positive by definition

Answer & Full Explanation**Correct Answer: (B)**

$T' + k\lambda^2 T = 0 \Rightarrow T = Ce^{-k\lambda^2 t}$. For $\lambda^2 > 0$: $T \rightarrow 0$ as $t \rightarrow \infty$ (**physically correct**: rod cools).

If we chose $+\lambda^2$: $T = Ce^{+k\lambda^2 t} \rightarrow \infty$ (unphysical blow-up). The negative sign for the separation constant is dictated by physical and mathematical (boundedness) requirements.

11. Important PSC Past-Paper Style MCQs — Mixed Topics

Fourier Series & PDEs Combined — BPS-17/PPSC/FPSC/CSS Standard

Q11.1. The Fourier series of $f(x) = \pi - x$ on $(0, 2\pi)$ (period 2π) is $2 \sum_{n=1}^{\infty} \frac{\sin nx}{n}$.

The sum at $x = \pi$ is:

- (A) π
- (B) 0
- (C) $\pi/2$
- (D) 1

Answer & Full Explanation**Correct Answer: (B) 0**

At $x = \pi$: $\sin(n\pi) = 0$ for all $n \in \mathbb{Z}$. So the series $= 2 \sum \frac{0}{n} = 0$.

By Dirichlet at $x = \pi$ (a point of continuity): the series equals $f(\pi) = \pi - \pi = 0$.
✓

Q11.2. The PDE $u_{xx} + u_{yy} + u_{zz} = 0$ is:

- (A) Parabolic in 3D
- (B) Hyperbolic
- (C) Elliptic — the 3D Laplace equation
- (D) Cannot be classified

Answer & Full Explanation

Correct Answer: (C) Elliptic

The 3D Laplace equation $\nabla^2 u = 0$ is elliptic. In 3D, the discriminant generalises: the coefficient matrix of second derivatives $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is positive definite $\Rightarrow$ elliptic.

Q11.3. By the Fourier cosine series of $f(x) = x^2$ on $(0, \pi)$, $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ equals:

- (A) $\frac{\pi^2}{6}$
- (B) $\frac{\pi^2}{12}$
- (C) $\frac{\pi^2}{4}$
- (D) 1

Answer & Full Explanation

Correct Answer: (B) $\frac{\pi^2}{12}$

Fourier cosine series of x^2 on $(0, \pi)$: $x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx$. At $x = 0$:

$$0 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}.$$

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\frac{\pi^2}{12} \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}.$$

Q11.4. The solution of Laplace's equation inside the unit disk $r < 1$ with $u(1, \theta) = \cos \theta$ is:

- (A) $u = r \cos \theta$
- (B) $u = \cos \theta / r$
- (C) $u = r^2 \cos \theta$
- (D) $u = r \sin \theta$

Answer & Full Explanation

Correct Answer: (A) $u = r \cos \theta$

$\nabla^2 u = 0$ in polar with $u(1, \theta) = \cos \theta$: Bounded solution in disk: $u =$

$$\sum_{n=0}^{\infty} r^n (a_n \cos n\theta + b_n \sin n\theta).$$

At $r = 1$: $u = \cos \theta \Rightarrow n = 1, a_1 = 1$, all others $= 0$. $u = r^1 \cos \theta = r \cos \theta$.

In Cartesian: $u = r \cos \theta = x$ (harmonic: $u_{xx} + u_{yy} = 0$). ✓

Q11.5. The wave equation $u_{tt} = c^2 u_{xx}$ with $u(x, 0) = 0$ and $u_t(x, 0) = g(x)$ (D'Alembert) gives:

- (A) $u = \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$
- (B) $u = \frac{1}{2} [g(x+ct) - g(x-ct)]$
- (C) $u = g(x-ct)$
- (D) $u = c g(x)t$

Answer & Full Explanation

Correct Answer: (A)

D'Alembert: $u = \frac{1}{2} [f(x+ct) + f(x-ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$. With $f = 0$:

$$u = \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds.$$

12. Quick-Revision Master Summary

Fourier Series & PDEs — Complete Formula Reference

Fourier coeff. a_n	$\frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx$
Fourier coeff. b_n	$\frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx$
Half-range sine b_n	$\frac{2}{L} \int_0^L f \sin \frac{n\pi x}{L} dx$
Half-range cosine a_n	$\frac{2}{L} \int_0^L f \cos \frac{n\pi x}{L} dx$
Parseval's identity	$\frac{a_0^2}{2} + \sum (a_n^2 + b_n^2) = \frac{1}{L} \int_{-L}^L f^2 dx$
Basel sums	$\sum 1/n^2 = \pi^2/6; \sum 1/n^4 = \pi^4/90; \sum 1/(2n-1)^2 = \pi^2/8$
Leibniz series	$\sum (-1)^n/(2n+1) = \pi/4$
Dirichlet convergence	Converges to f at continuity, to average at jump
Gibbs phenomenon	$\approx 9\%$ overshoot near jump; persists as $N \rightarrow \infty$
PDE discriminant	$\Delta = B^2 - 4AC: < 0$ elliptic, $= 0$ parabolic, > 0 hyperbolic
Laplace eq.	$u_{xx} + u_{yy} = 0$ (elliptic, steady-state); max on boundary
Wave eq.	$u_{tt} = c^2 u_{xx}$ (hyperbolic); $\omega_n = n\pi c/L$
Wave solution	$\sum \sin \frac{n\pi x}{L} (A_n \cos \omega_n t + B_n \sin \omega_n t)$
D'Alembert	$u = \frac{1}{2} [f(x+ct) + f(x-ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g ds$
Heat eq.	$u_t = k u_{xx}$ (parabolic); $T_n = e^{-k(n\pi/L)^2 t}$
Heat solution	$\sum B_n e^{-k(n\pi/L)^2 t} \sin \frac{n\pi x}{L}$
Homog. BC: $u \rightarrow 0$; Insulated: $u \rightarrow \frac{1}{L} \int_0^L f dx$	

PSC Exam Tip

Top 15 BPS-17/PSC Exam Strategies — Fourier Series & PDEs (Zamir Hussain):

1. **Even/Odd check first:** even $f \Rightarrow b_n = 0$ (cosine series); odd $f \Rightarrow a_n = 0$ (sine series). Saves half the computation.
2. **Parseval $\rightarrow$ series sums:** apply Parseval to $f(x) = x$ to get $\sum 1/n^2 = \pi^2/6$; to $f(x) = x^2$ to get $\sum 1/n^4 = \pi^4/90$.
3. **Dirichlet convergence:** at discontinuity, series converges to **average** — tested on $x = \pm\pi$ for $f(x) = x$.
4. **Gibbs phenomenon:** overshoot $\approx 9\%$, does not vanish as $N \rightarrow \infty$.
5. **Leibniz series:** $\sum (-1)^n/(2n+1) = \pi/4$ from square wave at $x = \pi/2$.
6. **Classification:** memorise $\Delta = B^2 - 4AC$; Laplace: elliptic, Heat: parabolic, Wave: hyperbolic.
7. **Sign of separation constant:** always $-\lambda^2$ in $X''/X = \dots$ so that X oscillates (sine/cosine).
8. **Heat vs Wave Y -equation:** Heat: $T' = -k\lambda^2 T$ (exponential decay); Wave: $T'' = -\lambda^2 c^2 T$ (oscillatory).
9. **Laplace rectangle:** Y satisfies $Y'' - \lambda^2 Y = 0$ (hyperbolic sines/cosines), opposite sign to X .
10. **Heat steady state:** zero BCs $\rightarrow u \rightarrow 0$; insulated BCs $\rightarrow u \rightarrow$ mean of IC; non-zero BCs $\rightarrow$ subtract linear steady state first.
11. **D'Alembert's formula:** works for infinite string; $f = 0, g \neq 0$: $u = \frac{1}{2c} \int_{x-ct}^{x+ct} g ds$.
12. **Maximum principle:** harmonic function attains max/min on boundary — implies uniqueness.
13. **Lagrange method:** write $dx/P = dy/Q = dz/R$; find two integrals; general solution $F(u_1, u_2) = 0$.
14. **Bessel inequality $\leq$ Parseval:** equality iff series converges in mean square (L^2).
15. **Natural frequencies:** $f_n = nc/(2L)$; fundamental $f_1 = c/(2L)$; overtones $= n \times f_1$.

Exam Practice Resource

For further exam practice, the following resource provides a comprehensive collection of solved MCQs:

[Access the material here](#)

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